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Education

University of Southern California
Ph.D. Economics — 2021-2027 (Expected)
Toulouse School of Economics
Visiting Ph.D. student — 2026
Hanyang University
M.A. Economics and Finance — 2020
B.S. Mathematics — 2018

Fields

Microeconomic Theory, Industrial Organization, Mechanism Design

References

Guofu Tan (Chair)
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Nicolas Lambert
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Jonathan Libgober
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Afshin Nikzad
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Job Market Paper

Consumer Learning and Competitive Inefficiency

We study price competition in a differentiated-product duopoly in which a consumer observes prices before deciding whether to acquire costly information about which product fits her better. Prices shape the value of learning and, in turn, the consumer's product choice. We characterize the unique mixed-strategy pricing equilibrium in which both firms are active. In equilibrium, sufficiently large price differences override product-fit information, generating ex post inefficiency even when information is free—a distortion absent under monopoly. How this inefficiency changes with information costs depends on market asymmetry: no learning maximizes total surplus when the allocation gain from favoring the advantaged firm outweighs the matching gain from learning. Otherwise, free information is optimal. Higher information costs initially reduce learning-induced product differentiation, intensify price competition, and raise consumer surplus. Once learning ceases, however, the advantaged firm captures the market and charges a higher price as information costs rise further, lowering consumer surplus. Finally, we endogenize the information friction and show that the firm may provide more product-fit transparency than is socially optimal.

Working Papers

Optimal Screening under Entry: Quantity Caps and Endogenous Market Segmentation *with Afshin Nikzad and Guofu Tan*

We study how entry reshapes monopoly screening under asymmetric information. In our model, an incumbent with full contractual flexibility moves first, followed by an entrant restricted to linear pricing. The firms compete to serve a downstream buyer with privately known willingness to pay and inelastic demand. Although the incumbent can offer an arbitrarily complex menu, its optimal mechanism consists of a single quantity-payment option. We establish this result in two steps. Starting from an arbitrary menu, we first construct a class of alternative menus that preserve the entrant's best response and use an extreme-point argument to show that the incumbent

can restrict attention to menus with at most two options. We then reduce any such two-option menu to a single option by reallocating quantity from lower- to higher-valuation types. The incumbent's optimal quantity-payment option can be implemented through a quantity cap and a per-unit price below the entrant's. The cap allows the incumbent to target intermediate-valuation types without competing directly for high-valuation types, who instead purchase from the entrant, thereby endogenously segmenting the market. Relative to the monopoly benchmark, entry reduces the incumbent's profit while increasing both consumer surplus and total surplus.

Predicting Merger Price Effects

A central concern in merger review is whether a proposed merger will raise prices and reduce consumer welfare. This paper examines how accurately the CPT-UPP method, proposed by Farrell and Shapiro (2010b), predicts merger price effects under logit demand. By combining upward pricing pressure with cost pass-through, CPT-UPP provides quantitative predictions using pre-merger information. We study mergers in which all firms combine into a single firm and fully characterize the prediction bias in symmetric equilibrium. When two firms merge, CPT-UPP overestimates the price effects below a unique threshold for each firm's pre-merger market share and underestimates them above it. When more than two firms merge, it consistently underestimates the price effects. We also conduct simulations for asymmetric duopolies to examine how the prediction bias varies with market-share asymmetry. These findings highlight the importance of market structure when interpreting CPT-UPP predictions in merger reviews.

Publications

Woo-Jin Kim and Youngwoo Koh. "Early Admissions with Multi-type Students." *Journal of Economic Research*, 26, 2021, pp. 209–236.

Work in Progress

Thickness and Competition in On-demand Service Platforms (with Afshin Nikzad)
Uniform vs Discriminatory Commissions in Platform Markets
Procurement with Quantity Guarantees: Equilibrium and Aftermarket Effects (with Yizhong Wang)

Awards

Department of Economics Dissertation Completion Fellowship, USC 2026
Summer Research Award, USC 2024, 2026
Professional development fund, Graduate Student Government 2024, 2026
Travel fund, USC 2024, 2026
Graduate School Fellowship, USC 2021-22, 2025-26
Study Abroad Scholarship, Hanyang Univ. 2021-23
Brain Korea 21 Plus Scholarship, Hanyang Univ. 2020-21
Graduate Student Scholarship, Korea Student Aid Foundation 2019

Research Positions

Research Assistant to Paulina Oliva, USC 2025
Research Assistant to Afshin Nikzad, USC 2022-23

Teaching

Instructor
Lectures for International Students (Math & Statistics), Hanyang Univ. 2018-19
Boot Camp for the Core Exam (Micro Session), USC 2023-24
Teaching Assistant
Microeconomic Theory I (Ph.D. core), USC Fall 2022-24
Microeconomic Theory II (Ph.D. core), USC Spring 2023-24

Antitrust Economics and Competition Policy (Master), USC Spring 2025

Presentations 2026: USC Theory Seminar (Scheduled), 37th Stony Brook International Conference on Game Theory, UCLA Student Theory Conference, TSE IO workshop, TSE IO Winter Workshop
2025: USC
2024: INFORMS (Seattle)

Software Skills Mathematica, MATLAB, Python

Languages English (fluent), Korean (native)

Citizenship South Korea (F-1 visa)